

DAY 5: MongoDB Assignment

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```
test> show dbs
admin          40.00 KiB
config         108.00 KiB
finance-tracker 40.00 KiB
local          76.00 KiB
test> use finance-tracker
switched to db finance-tracker
```

```
finance-tracker> show collections
```

```
transactions
```

```
finance-tracker> db.transactions.insertMany([
  { date: ISODate('2026-01-03T09:00:00Z'), category: 'Revenue', description: 'Freelance Project', amount: 2500, currency: 'USD', status: 'Completed' },
  { date: ISODate('2026-01-03T10:30:00Z'), category: 'Expense', description: 'Electricity Bill', amount: 85.50, currency: 'USD', status: 'Completed' },
  { date: ISODate('2026-01-04T11:00:00Z'), category: 'Expense', description: 'Internet Subscription', amount: 60, currency: 'USD', status: 'Completed' },
  { date: ISODate('2026-01-04T13:45:00Z'), category: 'Revenue', description: 'Stock Dividends', amount: 450.25, currency: 'USD', status: 'Completed' },
  { date: ISODate('2026-01-05T15:20:00Z'), category: 'Expense', description: 'New Office Chair', amount: 210, currency: 'USD', status: 'Completed' },
  { date: ISODate('2026-01-05T16:00:00Z'), category: 'Expense', description: 'Software License', amount: 300, currency: 'USD', status: 'Pending' },
  { date: ISODate('2026-01-06T08:15:00Z'), category: 'Revenue', description: 'Product Sale', amount: 120, currency: 'USD', status: 'Completed' },
  { date: ISODate('2026-01-06T12:00:00Z'), category: 'Expense', description: 'Team Lunch', amount: 75.40, currency: 'USD', status: 'Completed' },
  { date: ISODate('2026-01-07T10:00:00Z'), category: 'Revenue', description: 'Consulting Fee', amount: 3000, currency: 'USD', status: 'Completed' },
  { date: ISODate('2026-01-07T14:30:00Z'), category: 'Expense', description: 'Marketing Ads', amount: 500, currency: 'USD', status: 'Completed' },
  { date: ISODate('2026-01-08T09:45:00Z'), category: 'Expense', description: 'Domain Renewal', amount: 15, currency: 'USD', status: 'Completed' },
  { date: ISODate('2026-01-08T11:15:00Z'), category: 'Revenue', description: 'Affiliate Commission', amount: 850, currency: 'USD', status: 'Completed' },
  { date: ISODate('2026-01-09T13:00:00Z'), category: 'Expense', description: 'Office Supplies', amount: 45.20, currency: 'USD', status: 'Completed' },
  { date: ISODate('2026-01-09T17:00:00Z'), category: 'Expense', description: 'Travel Expenses', amount: 1200, currency: 'USD', status: 'Pending' },
  { date: ISODate('2026-01-10T10:30:00Z'), category: 'Revenue', description: 'Workshop Ticket', amount: 200, currency: 'USD', status: 'Completed' },
  { date: ISODate('2026-01-10T12:45:00Z'), category: 'Expense', description: 'Water Bill', amount: 30.15, currency: 'USD', status: 'Completed' },
])
```

```
{ date: ISODate('2026-01-11T09:00:00Z'), category: 'Revenue', description: 'App Store Sales', amount: 1500, currency: 'USD', status: 'Completed' },
{ date: ISODate('2026-01-11T15:30:00Z'), category: 'Expense', description: 'Insurance', amount: 400, currency: 'USD', status: 'Completed' },
{ date: ISODate('2026-01-12T11:00:00Z'), category: 'Revenue', description: 'Client Retainer', amount: 2000, currency: 'USD', status: 'Completed' },
{ date: ISODate('2026-01-12T14:00:00Z'), category: 'Expense', description: 'Courier Services', amount: 25, currency: 'USD', status: 'Completed' }
])
```

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6957540bf32766074ccc8988'),
    '1': ObjectId('6957540bf32766074ccc8989'),
    '2': ObjectId('6957540bf32766074ccc898a'),
    '3': ObjectId('6957540bf32766074ccc898b'),
    '4': ObjectId('6957540bf32766074ccc898c'),
    '5': ObjectId('6957540bf32766074ccc898d'),
    '6': ObjectId('6957540bf32766074ccc898e'),
    '7': ObjectId('6957540bf32766074ccc898f'),
    '8': ObjectId('6957540bf32766074ccc8990'),
    '9': ObjectId('6957540bf32766074ccc8991'),
    '10': ObjectId('6957540bf32766074ccc8992'),
    '11': ObjectId('6957540bf32766074ccc8993'),
    '12': ObjectId('6957540bf32766074ccc8994'),
    '13': ObjectId('6957540bf32766074ccc8995'),
    '14': ObjectId('6957540bf32766074ccc8996'),
    '15': ObjectId('6957540bf32766074ccc8997'),
    '16': ObjectId('6957540bf32766074ccc8998'),
    '17': ObjectId('6957540bf32766074ccc8999'),
    '18': ObjectId('6957540bf32766074ccc899a'),
    '19': ObjectId('6957540bf32766074ccc899b')
  }
}
```

1. Print all documents in a collection?

db.transactions.find()

```
finance-tracker> show collections
transactions
finance-tracker> db.transactions.find()
[
  {
    _id: ObjectId('69574b50f83b29de13aa7b34'),
    date: ISODate('2026-01-02T10:00:00.000Z'),
    category: 'Revenue',
    description: 'Consulting Fee',
    amount: 5000,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('69574b50f83b29de13aa7b35'),
    date: ISODate('2026-01-02T11:30:00.000Z'),
    category: 'Expense',
    description: 'Office Rent',
    amount: 1200,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('69574b50f83b29de13aa7b36'),
    date: ISODate('2026-01-02T14:15:00.000Z'),
    category: 'Expense',
    description: 'Cloud Hosting',
    amount: 150.75,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc8988'),
    date: ISODate('2026-01-03T09:00:00.000Z'),
    category: 'Revenue',
    description: 'Freelance Project',
    amount: 2500,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc8989'),
    date: ISODate('2026-01-03T10:30:00.000Z'),
    category: 'Expense',
    description: 'Electricity Bill',
    amount: 85.5,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc898a'),
    date: ISODate('2026-01-04T11:00:00.000Z'),
    category: 'Expense',
    description: 'Internet Subscription',
    amount: 60,
    currency: 'USD',
    status: 'Completed'
  }
]
```

2. Sort transactions by amount (highest first) and show top 10?

`db.transactions.find().sort({ amount: -1 }).limit(10)`

```
finance-tracker> db.transactions.find().sort({ amount: -1 }).limit(10)
[
  {
    _id: ObjectId('69574b50f83b29de13aa7b34'),
    date: ISODate('2026-01-02T10:00:00.000Z'),
    category: 'Revenue',
    description: 'Consulting Fee',
    amount: 5000,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc8990'),
    date: ISODate('2026-01-07T10:00:00.000Z'),
    category: 'Revenue',
    description: 'Consulting Fee',
    amount: 3000,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc8988'),
    date: ISODate('2026-01-03T09:00:00.000Z'),
    category: 'Revenue',
    description: 'Freelance Project',
    amount: 2500,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc899a'),
    date: ISODate('2026-01-12T11:00:00.000Z'),
    category: 'Revenue',
    description: 'Client Retainer',
    amount: 2000,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc8998'),
    date: ISODate('2026-01-11T09:00:00.000Z'),
    category: 'Revenue',
    description: 'App Store Sales',
    amount: 1500,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('69574b50f83b29de13aa7b35'),
    date: ISODate('2026-01-02T11:30:00.000Z'),
    category: 'Expense',
    description: 'Office Rent',
    amount: 1200,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc8993'),
    date: ISODate('2026-01-08T11:15:00.000Z'),
    category: 'Revenue',
    description: 'Affiliate Commission',
    amount: 850,
    currency: 'USD',
    status: 'Completed'
  }
]
```

3. Sort transactions by date (oldest first) and limit results?

`db.transactions.find().sort({ date: 1 }).limit(3)`

```
finance-tracker> db.transactions.find().sort({ date: 1 }).limit(3)
[
  {
    _id: ObjectId('69574b50f83b29de13aa7b34'),
    date: ISODate('2026-01-02T10:00:00.000Z'),
    category: 'Revenue',
    description: 'Consulting Fee',
    amount: 5000,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('69574b50f83b29de13aa7b35'),
    date: ISODate('2026-01-02T11:30:00.000Z'),
    category: 'Expense',
    description: 'Office Rent',
    amount: 1200,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('69574b50f83b29de13aa7b36'),
    date: ISODate('2026-01-02T14:15:00.000Z'),
    category: 'Expense',
    description: 'Cloud Hosting',
    amount: 150.75,
    currency: 'USD',
    status: 'Completed'
  }
]
finance-tracker> |
```

4. Get top 5 highest expense transactions?

`db.transactions.find({ category: "Expense" }).sort({ amount: -1 }).limit(5)`

```
finance-tracker> db.transactions.find({ category: "Expense" }).sort({ amount: -1 }).limit(5)
[
  {
    _id: ObjectId('69574b50f83b29de13aa7b35'),
    date: ISODate('2026-01-02T11:30:00.000Z'),
    category: 'Expense',
    description: 'Office Rent',
    amount: 1200,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('69575a0bf32766074ccc8991'),
    date: ISODate('2026-01-07T14:30:00.000Z'),
    category: 'Expense',
    description: 'Marketing Ads',
    amount: 500,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('69575a0bf32766074ccc8995'),
    date: ISODate('2026-01-11T15:30:00.000Z'),
    category: 'Expense',
    description: 'Insurance',
    amount: 400,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('69575a0bf32766074ccc899c'),
    date: ISODate('2026-01-05T15:20:00.000Z'),
    category: 'Expense',
    description: 'New Office Chair',
    amount: 210,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('69574b50f83b29de13aa7b36'),
    date: ISODate('2026-01-02T14:15:00.000Z'),
    category: 'Expense',
    description: 'Cloud Hosting',
    amount: 150.75,
    currency: 'USD',
    status: 'Completed'
  }
]
finance-tracker> |
```

5. Calculate total amount and count per category (Revenue vs Expense)?

```
db.transactions.aggregate([
  {
    $group: {
      _id: "$category",
      totalAmount: { $sum: "$amount" },
      count: { $sum: 1 }
    }
  }
])
```

```
finance-tracker> db.transactions.aggregate([ { $group: { _id: "$category", totalAmount: { $sum: "$amount" }, count: { $sum: 1 } } } ] )
[
  { _id: 'Expense', totalAmount: 2797, count: 12 },
  { _id: 'Revenue', totalAmount: 15620.25, count: 9 }
]
finance-tracker> |
```

6. Update a specific transaction's status?

```
db.transactions.updateOne(
  { description: "Office Rent" },
  { $set: { status: "Completed" } }
)
```

```
finance-tracker> db.transactions.updateOne(
...   { description: "Office Rent" },
...   { $set: { status: "Completed" } }
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 0,
  upsertedCount: 0
}
finance-tracker> |
```

7. Find transactions between two dates?

```
db.transactions.find({
  date: {
    $gte: ISODate("2026-01-01T00:00:00Z"),
    $lte: ISODate("2026-01-05T23:59:59Z")
  }
})
```

```
finance-tracker> db.transactions.find({
...   date: {
...     $gte: ISODate("2026-01-01T00:00:00Z"),
...     $lte: ISODate("2026-01-05T23:59:59Z")
...   }
... })
[
  {
    _id: ObjectId('69574b50f83b29de13aa7b34'),
    date: ISODate('2026-01-02T10:00:00.000Z'),
    category: 'Revenue',
    description: 'Consulting Fee',
    amount: 5000,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('69574b50f83b29de13aa7b35'),
    date: ISODate('2026-01-02T11:30:00.000Z'),
    category: 'Expense',
    description: 'Office Rent',
    amount: 1200,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('69574b50f83b29de13aa7b36'),
    date: ISODate('2026-01-02T14:15:00.000Z'),
    category: 'Expense',
    description: 'Cloud Hosting',
    amount: 150.75,
    currency: 'USD',
    status: 'Completed'
  }
],
```

8. Find transactions with amount greater than 1000?

db.transactions.find({ amount: { \$gt: 1000 } })

```
finance-tracker> db.transactions.find({ amount: { $gt: 1000 } })
[
  {
    _id: ObjectId('69574b50f83b29de13aa7b34'),
    date: ISODate('2026-01-02T10:00:00.000Z'),
    category: 'Revenue',
    description: 'Consulting Fee',
    amount: 5000,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('69574b50f83b29de13aa7b35'),
    date: ISODate('2026-01-02T11:30:00.000Z'),
    category: 'Expense',
    description: 'Office Rent',
    amount: 1200,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc8988'),
    date: ISODate('2026-01-03T09:00:00.000Z'),
    category: 'Revenue',
    description: 'Freelance Project',
    amount: 2500,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc8990'),
    date: ISODate('2026-01-07T10:00:00.000Z'),
    category: 'Revenue',
    description: 'Consulting Fee',
    amount: 3000,
    currency: 'USD',
    status: 'Completed'
  }
],
```

9.Delete all pending transactions?

```
db.transactions.deleteMany({ status: "Pending" })
```

```
finance-tracker> db.transactions.deleteMany({ status: "Pending" })
{ acknowledged: true, deletedCount: 0 }
finance-tracker> |
```

10. Count total documents in a collection?

```
db.transactions.countDocuments({})
```

```
finance-tracker> db.transactions.countDocuments({})
21
finance-tracker> |
```

11. Filter only Expense transactions?

```
db.transactions.find({ category: "Expense" })
```

```
finance-tracker> db.transactions.find({ category: "Expense" })
[
  {
    _id: ObjectId('69574b50f83b29de13aa7b35'),
    date: ISODate('2026-01-02T11:30:00.000Z'),
    category: 'Expense',
    description: 'Office Rent',
    amount: 1200,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('69574b50f83b29de13aa7b36'),
    date: ISODate('2026-01-02T14:15:00.000Z'),
    category: 'Expense',
    description: 'Cloud Hosting',
    amount: 150.75,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc8989'),
    date: ISODate('2026-01-03T10:30:00.000Z'),
    category: 'Expense',
    description: 'Electricity Bill',
    amount: 85.5,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc898a'),
    date: ISODate('2026-01-04T11:00:00.000Z'),
    category: 'Expense',
    description: 'Internet Subscription',
    amount: 60,
    currency: 'USD',
    status: 'Completed'
  },
]
```


12. Filter only Revenue transactions?

`db.transactions.find({ category: "Revenue" })`

```
finance-tracker> db.transactions.find({ category: "Revenue" })
[
  {
    _id: ObjectId('69574b50f83b29de13aa7b34'),
    date: ISODate('2026-01-02T10:00:00.000Z'),
    category: 'Revenue',
    description: 'Consulting Fee',
    amount: 5000,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc8988'),
    date: ISODate('2026-01-03T09:00:00.000Z'),
    category: 'Revenue',
    description: 'Freelance Project',
    amount: 2500,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc898b'),
    date: ISODate('2026-01-04T13:45:00.000Z'),
    category: 'Revenue',
    description: 'Stock Dividends',
    amount: 450.25,
    currency: 'USD',
    status: 'Completed'
  },
]
```

13. Sort transactions by date (latest first)?

`db.transactions.find().sort({ date: -1 })`

```
finance-tracker> db.transactions.find().sort({ date: -1 })
[
  {
    _id: ObjectId('6957540bf32766074ccc899b'),
    date: ISODate('2026-01-12T14:00:00.000Z'),
    category: 'Expense',
    description: 'Courier Services',
    amount: 25,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc899a'),
    date: ISODate('2026-01-12T11:00:00.000Z'),
    category: 'Revenue',
    description: 'Client Retainer',
    amount: 2000,
    currency: 'USD',
    status: 'Completed'
  },
]
```

14. Find completed transactions only?

```
db.transactions.find({ status: "Completed" })
```

```
finance-tracker> db.transactions.find({ status: "Completed" })
[
  {
    _id: ObjectId('69574b50f83b29de13aa7b34'),
    date: ISODate('2026-01-02T10:00:00.000Z'),
    category: 'Revenue',
    description: 'Consulting Fee',
    amount: 5000,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('69574b50f83b29de13aa7b35'),
    date: ISODate('2026-01-02T11:30:00.000Z'),
    category: 'Expense',
    description: 'Office Rent',
    amount: 1200,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('69574b50f83b29de13aa7b36'),
    date: ISODate('2026-01-02T14:15:00.000Z'),
    category: 'Expense',
    description: 'Cloud Hosting',
    amount: 150.75,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc8988'),
    date: ISODate('2026-01-03T09:00:00.000Z'),
    category: 'Revenue',
    description: 'Freelance Project',
    amount: 2500,
    currency: 'USD',
    status: 'Completed'
  }
]
```

15. Combine filter + sort (Expense, highest amount)?

```
db.transactions.find({ category: "Expense" }).sort({ amount: -1 })
```

```
finance-tracker> db.transactions.find({ category: "Expense" }).sort({ amount: -1 })
[
  {
    _id: ObjectId('69574b50f83b29de13aa7b35'),
    date: ISODate('2026-01-02T11:30:00.000Z'),
    category: 'Expense',
    description: 'Office Rent',
    amount: 1200,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc8991'),
    date: ISODate('2026-01-07T14:30:00.000Z'),
    category: 'Expense',
    description: 'Marketing Ads',
    amount: 500,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc8999'),
    date: ISODate('2026-01-11T15:30:00.000Z'),
    category: 'Expense',
    description: 'Insurance',
    amount: 400,
    currency: 'USD',
    status: 'Completed'
  }
]
```

16. Find documents with amount less than 500?

`db.transactions.find({ amount: { $lt: 500 } })`

```
finance-tracker> db.transactions.find({ amount: { $lt: 500 } })
[
  {
    _id: ObjectId('69574b50f83b29de13aa7b36'),
    date: ISODate('2026-01-02T14:15:00.000Z'),
    category: 'Expense',
    description: 'Cloud Hosting',
    amount: 150.75,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc8989'),
    date: ISODate('2026-01-03T10:30:00.000Z'),
    category: 'Expense',
    description: 'Electricity Bill',
    amount: 85.5,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc898a'),
    date: ISODate('2026-01-04T11:00:00.000Z'),
    category: 'Expense',
    description: 'Internet Subscription',
    amount: 60,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc898b'),
    date: ISODate('2026-01-04T13:45:00.000Z'),
    category: 'Revenue',
    description: 'Stock Dividends',
    amount: 450.25,
    currency: 'USD',
    status: 'Completed'
  }
]
```

17. Find documents with amount less than or equal to 300?

`db.transactions.find({ amount: { $lte: 300 } })`

```
finance-tracker> db.transactions.find({ amount: { $lte: 300 } })
[
  {
    _id: ObjectId('69574b50f83b29de13aa7b36'),
    date: ISODate('2026-01-02T14:15:00.000Z'),
    category: 'Expense',
    description: 'Cloud Hosting',
    amount: 150.75,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc8989'),
    date: ISODate('2026-01-03T10:30:00.000Z'),
    category: 'Expense',
    description: 'Electricity Bill',
    amount: 85.5,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc898a'),
    date: ISODate('2026-01-04T11:00:00.000Z'),
    category: 'Expense',
    description: 'Internet Subscription',
    amount: 60,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc898c'),
    date: ISODate('2026-01-05T15:20:00.000Z'),
    category: 'Expense',
    description: 'New Office Chair',
    amount: 210,
    currency: 'USD',
    status: 'Completed'
  }
]
```

18. Find documents where category is NOT Expense?

```
db.transactions.find({ category: { $ne: "Expense" } })
```

```
finance-tracker> db.transactions.find({ category: { $ne: "Expense" } })
[
  {
    _id: ObjectId('69574b50f83b29de13aa7b34'),
    date: ISODate('2026-01-02T10:00:00.000Z'),
    category: 'Revenue',
    description: 'Consulting Fee',
    amount: 5000,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc8988'),
    date: ISODate('2026-01-03T09:00:00.000Z'),
    category: 'Revenue',
    description: 'Freelance Project',
    amount: 2500,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc898b'),
    date: ISODate('2026-01-04T13:45:00.000Z'),
    category: 'Revenue',
    description: 'Stock Dividends',
    amount: 450.25,
    currency: 'USD',
    status: 'Completed'
  }
]
```

19. Find transactions with specific descriptions using \$in?

```
db.transactions.find({
  description: { $in: ["Office Rent", "Marketing Ads"] }
})
```

```
finance-tracker> db.transactions.find({
...   description: { $in: ["Office Rent", "Marketing Ads"] }
... })
[
  {
    _id: ObjectId('69574b50f83b29de13aa7b35'),
    date: ISODate('2026-01-02T11:30:00.000Z'),
    category: 'Expense',
    description: 'Office Rent',
    amount: 1200,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc8991'),
    date: ISODate('2026-01-07T14:30:00.000Z'),
    category: 'Expense',
    description: 'Marketing Ads',
    amount: 500,
    currency: 'USD',
    status: 'Completed'
  }
]
finance-tracker>
```

20. Find documents where a field does NOT exist?

db.transactions.find({ verified: { \$exists: false } })

```
finance-tracker> db.transactions.find({ verified: { $exists: false } })
[
  {
    _id: ObjectId('69574b50f83b29de13aa7b34'),
    date: ISODate('2026-01-02T10:00:00.000Z'),
    category: 'Revenue',
    description: 'Consulting Fee',
    amount: 5000,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('69574b50f83b29de13aa7b35'),
    date: ISODate('2026-01-02T11:30:00.000Z'),
    category: 'Expense',
    description: 'Office Rent',
    amount: 1200,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('69574b50f83b29de13aa7b36'),
    date: ISODate('2026-01-02T14:15:00.000Z'),
    category: 'Expense',
    description: 'Cloud Hosting',
    amount: 150.75,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc8988'),
    date: ISODate('2026-01-03T09:00:00.000Z'),
    category: 'Revenue',
    description: 'Freelance Project',
    amount: 2500,
    currency: 'USD',
    status: 'Completed'
  }
]
```

21. Sort transactions by date (latest first) and limit to 3?

db.transactions.find().sort({ date: -1 }).limit(3)

```
finance-tracker> db.transactions.find().sort({ date: -1 }).limit(3)
[
  {
    _id: ObjectId('6957540bf32766074ccc899b'),
    date: ISODate('2026-01-12T14:00:00.000Z'),
    category: 'Expense',
    description: 'Courier Services',
    amount: 25,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc899a'),
    date: ISODate('2026-01-12T11:00:00.000Z'),
    category: 'Revenue',
    description: 'Client Retainer',
    amount: 2000,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc8999'),
    date: ISODate('2026-01-11T15:30:00.000Z'),
    category: 'Expense',
    description: 'Insurance',
    amount: 400,
    currency: 'USD',
    status: 'Completed'
  }
]
finance-tracker>
```

22. Skip first 5 documents and show next 3?

db.transactions.find().skip(5).limit(3)

```
finance-tracker> db.transactions.find().skip(5).limit(3)
[
  {
    _id: ObjectId('6957540bf32766074ccc898a'),
    date: ISODate('2026-01-04T11:00:00.000Z'),
    category: 'Expense',
    description: 'Internet Subscription',
    amount: 60,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc898b'),
    date: ISODate('2026-01-04T13:45:00.000Z'),
    category: 'Revenue',
    description: 'Stock Dividends',
    amount: 450.25,
    currency: 'USD',
    status: 'Completed'
  },
  {
    _id: ObjectId('6957540bf32766074ccc898c'),
    date: ISODate('2026-01-05T15:20:00.000Z'),
    category: 'Expense',
    description: 'New Office Chair',
    amount: 210,
    currency: 'USD',
    status: 'Completed'
  }
]
finance-tracker>
```

23. Display only description and status fields?

```
db.transactions.find(
  {},
  { description: 1, status: 1, _id: 0 }
)
```

```
finance-tracker> db.transactions.find(
...  {},
...  { description: 1, status: 1, _id: 0 }
... )
[
  { description: 'Consulting Fee', status: 'Completed' },
  { description: 'Office Rent', status: 'Completed' },
  { description: 'Cloud Hosting', status: 'Completed' },
  { description: 'Freelance Project', status: 'Completed' },
  { description: 'Electricity Bill', status: 'Completed' },
  { description: 'Internet Subscription', status: 'Completed' },
  { description: 'Stock Dividends', status: 'Completed' },
  { description: 'New Office Chair', status: 'Completed' },
  { description: 'Product Sale', status: 'Completed' },
  { description: 'Team Lunch', status: 'Completed' },
  { description: 'Consulting Fee', status: 'Completed' },
  { description: 'Marketing Ads', status: 'Completed' },
  { description: 'Domain Renewal', status: 'Completed' },
  { description: 'Affiliate Commission', status: 'Completed' },
  { description: 'Office Supplies', status: 'Completed' },
  { description: 'Workshop Ticket', status: 'Completed' },
  { description: 'Water Bill', status: 'Completed' },
  { description: 'App Store Sales', status: 'Completed' },
  { description: 'Insurance', status: 'Completed' },
  { description: 'Client Retainer', status: 'Completed' }
]
Type "it" for more
finance-tracker> |
```

24. Rename a field using projection?

```
db.transactions.find(
  {},
  { _id: 0, info: "$description", amount: 1 }
)
```

```
finance-tracker> db.transactions.find(
...   {},
...   { _id: 0, info: "$description", amount: 1 }
... )
...
[
  { amount: 5000, info: 'Consulting Fee' },
  { amount: 1200, info: 'Office Rent' },
  { amount: 150.75, info: 'Cloud Hosting' },
  { amount: 2500, info: 'Freelance Project' },
  { amount: 85.5, info: 'Electricity Bill' },
  { amount: 60, info: 'Internet Subscription' },
  { amount: 450.25, info: 'Stock Dividends' },
  { amount: 210, info: 'New Office Chair' },
  { amount: 120, info: 'Product Sale' },
  { amount: 75.4, info: 'Team Lunch' },
  { amount: 3000, info: 'Consulting Fee' },
  { amount: 500, info: 'Marketing Ads' },
  { amount: 15, info: 'Domain Renewal' },
  { amount: 850, info: 'Affiliate Commission' },
  { amount: 45.2, info: 'Office Supplies' },
  { amount: 200, info: 'Workshop Ticket' },
  { amount: 30.15, info: 'Water Bill' },
  { amount: 1500, info: 'App Store Sales' },
  { amount: 400, info: 'Insurance' },
  { amount: 2000, info: 'Client Retainer' }
]
Type "it" for more
finance-tracker>
```

25. Count only completed transactions?

```
db.transactions.countDocuments({ status: "Completed" })
```

```
finance-tracker> db.transactions.countDocuments({ status: "Completed" })
21
finance-tracker>
```

26. Update status for multiple Expense records?

```
db.transactions.updateMany(
  { category: "Expense" },
  { $set: { reviewed: true } }
)
```

```
finance-tracker> db.transactions.updateMany(
...   { category: "Expense" },
...   { $set: { reviewed: true } }
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 12,
  modifiedCount: 12,
  upsertedCount: 0
}
finance-tracker>
```

27. Remove a field from all documents?

```
db.transactions.updateMany(  
  {},  
  { $unset: { reviewed: "" } }  
)
```

```
finance-tracker> db.transactions.updateMany(  
...   {},  
...   { $unset: { reviewed: "" } }  
... )  
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 21,  
  modifiedCount: 12,  
  upsertedCount: 0  
}
```

28. Increase all Revenue amounts by 10%?

```
db.transactions.updateMany(  
  { category: "Revenue" },  
  { $mul: { amount: 1.1 } }  
)
```

```
finance-tracker> db.transactions.updateMany(  
...   { category: "Revenue" },  
...   { $mul: { amount: 1.1 } }  
... )  
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 9,  
  modifiedCount: 9,  
  upsertedCount: 0  
}
```

29. Delete one transaction using description?

```
db.transactions.deleteOne({ description: "Water Bill" })
```

```
finance-tracker> db.transactions.deleteOne({ description: "Water Bill" })  
{ acknowledged: true, deletedCount: 1 }  
finance-tracker>
```

30. Delete transactions with amount less than 50?

```
db.transactions.deleteMany({ amount: { $lt: 50 } })
```

```
finance-tracker> db.transactions.deleteMany({ amount: { $lt: 50 } })  
{ acknowledged: true, deletedCount: 3 }  
finance-tracker>
```


31. Calculate total amount per status using aggregation?

```
db.transactions.aggregate([
  { $group: { _id: "$status", total: { $sum: "$amount" } } }
])
```

```
finance-tracker> db.transactions.aggregate([
...   { $group: { _id: "$status", total: { $sum: "$amount" } } }
... ])
[ { _id: 'Completed', total: 19863.925000000003 } ]
finance-tracker>
```

32. Calculate average Expense amount only?

```
db.transactions.aggregate([
  { $match: { category: "Expense" } },
  { $group: { _id: null, avgExpense: { $avg: "$amount" } } }
])
```

```
finance-tracker> db.transactions.aggregate([
...   { $match: { category: "Expense" } },
...   { $group: { _id: null, avgExpense: { $avg: "$amount" } } }
... ])
[ { _id: null, avgExpense: 335.20625 } ]
finance-tracker>
```

33. Find maximum transaction amount per category?

```
db.transactions.aggregate([
  { $group: { _id: "$category", maxAmount: { $max: "$amount" } } }
])
```

```
finance-tracker> db.transactions.aggregate([
...   { $group: { _id: "$category", maxAmount: { $max: "$amount" } } }
... ])
[
  { _id: 'Revenue', maxAmount: 5500 },
  { _id: 'Expense', maxAmount: 1200 }
]
finance-tracker>
```

34. Convert category text to uppercase in output?

```
db.transactions.aggregate([
```

```
{
  $project: {
    _id: 0,
    categoryUpper: { $toUpper: "$category" },
    amount: 1
  }
}]
```

```
finance-tracker> db.transactions.aggregate([
...   {
...     $project: {
...       _id: 0,
...       categoryUpper: { $toUpper: "$category" },
...       amount: 1
...     }
...   }
... ])
[
  { amount: 5500, categoryUpper: 'REVENUE' },
  { amount: 1200, categoryUpper: 'EXPENSE' },
  { amount: 150.75, categoryUpper: 'EXPENSE' },
  { amount: 2750, categoryUpper: 'REVENUE' },
  { amount: 85.5, categoryUpper: 'EXPENSE' },
  { amount: 60, categoryUpper: 'EXPENSE' },
  { amount: 495.27500000000003, categoryUpper: 'REVENUE' },
  { amount: 210, categoryUpper: 'EXPENSE' },
  { amount: 132, categoryUpper: 'REVENUE' },
  { amount: 75.4, categoryUpper: 'EXPENSE' },
  { amount: 3300.0000000000005, categoryUpper: 'REVENUE' },
  { amount: 500, categoryUpper: 'EXPENSE' },
  { amount: 935.0000000000001, categoryUpper: 'REVENUE' },
  { amount: 220.00000000000003, categoryUpper: 'REVENUE' },
  { amount: 1650.0000000000002, categoryUpper: 'REVENUE' },
  { amount: 400, categoryUpper: 'EXPENSE' },
  { amount: 2200, categoryUpper: 'REVENUE' }
]
finance-tracker> |
```

35. Classify transactions as High or Low using \$cond?

```
db.transactions.aggregate([
```

```
{
  $project: {
    amount: 1,
    level: {
      $cond: [{ $gt: ["$amount", 1000] }, "High", "Low"]
    }
  }
}]
```

```

finance-tracker> db.transactions.aggregate([
...   {
...     $project: {
...       amount: 1,
...       level: {
...         $cond: [{ $gt: ["$amount", 1000] }, "High", "Low"]
...       }
...     }
...   }
... ])
[
  {
    _id: ObjectId('69574b50f83b29de13aa7b34'),
    amount: 5500,
    level: 'High'
  },
  {
    _id: ObjectId('69574b50f83b29de13aa7b35'),
    amount: 1200,
    level: 'High'
  },
  {
    _id: ObjectId('69574b50f83b29de13aa7b36'),
    amount: 150.75,
    level: 'Low'
  },
  {
    _id: ObjectId('6957540bf32766074ccc8988'),
    amount: 2750,
    level: 'High'
  },
]

```